

WEEK-7

Task-1: Write a program that prints the addresses Of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Code:

```
pavan@DESKTOP-38OLEIN: ~/practical
GNU nano 8.7.1 address.c *
#include <stdio.h>
#include <stdlib.h>

int global_var = 10;

void code_function()
{
    printf("Code address    : %p\n", (void *)code_function);
}

int main()
{
    static int static_var = 20;
    int stack_var = 30;
    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    *heap_var = 40;

    printf("Memory Addresses:\n");

    code_function();

    printf("Global address : %p\n", (void *)&global_var);
    printf("Static address : %p\n", (void *)&static_var);
    printf("Heap address   : %p\n", (void *)heap_var);
    printf("Stack address  : %p\n", (void *)&stack_var);

    free(heap_var);

    return 0;
}
```

Output:

```
pavani@DESKTOP-380LEIN:~$ mkdir practical
pavani@DESKTOP-380LEIN:~$ cd practical
pavani@DESKTOP-380LEIN:~/practical$ nano address.c
pavani@DESKTOP-380LEIN:~/practical$ gcc address.c -o address
pavani@DESKTOP-380LEIN:~/practical$ ./address
Memory Addresses:
Code address   : 0x5c14f8dfa1c9
Global address : 0x5c14f8dfd010
Static address : 0x5c14f8dfd014
Heap address   : 0x5c1528d57010
Stack address  : 0x7ffde11824ac
pavani@DESKTOP-380LEIN:~/practical$
```

Report: The program was successfully implemented to display addresses of different memory segments.

Addresses of code, global, static, heap, and stack variables were observed.

The results showed that different types of data are stored in different regions of virtual memory. Linux uses virtual address space and address randomization, so the exact addresses may change between executions.

Task-2: Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

Code:

```

pavani@DESKTOP-38OLEIN: ~/practical
GNU nano 8.7.1 maps.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Process ID: %d\n", getpid());
    printf("Process is running...\n");
    printf("Check /proc/%d/maps from another terminal.\n", getpid());

    while (1)
    {
        sleep(5);
    }

    return 0;
}

```

Output:

```

pavani@DESKTOP-38OLEIN:~$ cat /proc/12345/maps
cat: /proc/12345/maps: No such file or directory
pavani@DESKTOP-38OLEIN:~$ cat /proc/687/maps
5e2dc2f05000-5e2dc2f06000 r--p 00000000 08:30 34759 /home/pavani/practical/maps
5e2dc2f06000-5e2dc2f07000 r-xp 00001000 08:30 34759 /home/pavani/practical/maps
5e2dc2f07000-5e2dc2f08000 r--p 00002000 08:30 34759 /home/pavani/practical/maps
5e2dc2f08000-5e2dc2f09000 r--p 00002000 08:30 34759 /home/pavani/practical/maps
5e2dc2f09000-5e2dc2f0a000 rw-p 00003000 08:30 34759 /home/pavani/practical/maps
5e2df041c000-5e2df943d000 rw-p 00000000 00:00 0 [heap]
72a87d200000-72a87d228000 r--p 00000000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
72a87d228000-72a87d3c0000 r-xp 00028000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
72a87d3c0000-72a87d40e000 r--p 001c0000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
72a87d40e000-72a87d412000 r--p 0020d000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
72a87d412000-72a87d414000 rw-p 00211000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
72a87d414000-72a87d421000 rw-p 00000000 00:00 0
72a87d484000-72a87d487000 rw-p 00000000 00:00 0
72a87d48c000-72a87d48e000 rw-p 00000000 00:00 0
72a87d48e000-72a87d492000 r--p 00000000 00:00 0 [vvar]
72a87d492000-72a87d494000 r--p 00000000 00:00 0 [vvar_vclock]
72a87d494000-72a87d496000 r-xp 00000000 00:00 0 [vdso]
72a87d496000-72a87d497000 r--p 00000000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
72a87d497000-72a87d4c6000 r-xp 00001000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
72a87d4c6000-72a87d4d1000 r--p 00030000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
72a87d4d1000-72a87d4d3000 r--p 0003b000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
72a87d4d3000-72a87d4d4000 rw-p 0003d000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
72a87d4d4000-72a87d4d5000 rw-p 00000000 00:00 0
7ffe993f1000-7ffe99412000 rw-p 00000000 00:00 0 [stack]
pavani@DESKTOP-38OLEIN:~$

```

Report: The `/proc/PID/maps` file was used to study the virtual memory layout of a running process.

Different mappings such as code, data, heap, stack, and shared libraries were observed.

The permissions of each region were analyzed using read, write, and execute flags.

This experiment showed how Linux organizes and manages a process using virtual memory mappings.